

Workbook: Cardiac Conduction

Week 3 workbook · BIO 004 Human Anatomy · Dr. Sharilyn Rennie · Solano Community College

A comprehensive workbook on the heart's intrinsic conduction system: the structures that generate and propagate the signal, the order in which the chambers contract, the basic ECG, and clinical anatomy of conduction problems.

How to use this workbook: Read each question carefully and write your answer in the space provided. Work without looking up the answer first. Once you have written your answer, check it against the online workbook by clicking *Reveal* on that question. Star or circle anything you missed and come back to it later in the week.

The conduction system structures · 6 questions

Q01 · DOK 1 · Recall

Name the components of the cardiac conduction system in order from where the signal starts to where it ends.

Q02 · DOK 1 · Recall

Where is the SA node, and what is its role?

Q03 · DOK 1 · Recall

Where is the AV node, and what is its key feature?

Q04 • DOK 1 • Recall

Where is the AV bundle (bundle of His), and what does it do?

Q05 • DOK 1 • Recall

What are Purkinje fibers, and what do they do?

Q06 • DOK 2 • Apply

Why do the ventricles begin contracting at the apex and squeeze upward toward the base?

The autonomic input • 2 questions**Q07 • DOK 1 • Recall**

How do the sympathetic and parasympathetic nervous systems influence the heart?

Q08 • DOK 2 • Apply

The heart will beat even when all autonomic nerves are cut. What does that tell you about the conduction system?

The ECG • 3 questions

Q09 • DOK 1 • Recall

Name the three waves of a normal ECG and what each represents.

Q10 • DOK 2 • Apply

Why does the ECG not show a separate wave for atrial repolarization?

Q11 • DOK 2 • Apply

What does the PR interval on an ECG represent, and what would lengthen it?

Clinical reasoning • 3 questions**Q12 • DOK 3 • Reason**

If the SA node fails, the heart usually keeps beating, but more slowly. Why?

In complete (third-degree) AV block, no signal gets from the atria to the ventricles. Predict what happens to the heartbeat.

A myocardial infarction affecting the interventricular septum can damage the bundle branches. Predict what kind of ECG abnormality might appear.